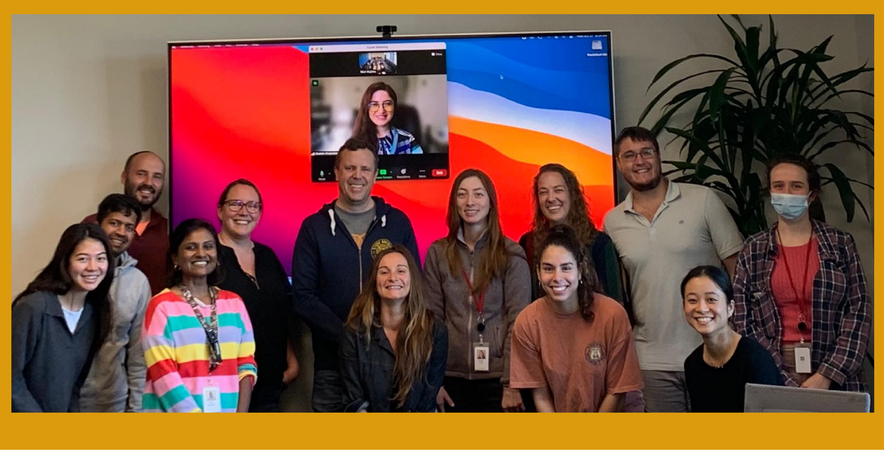

A series of formal contests to predict *B. pertussis* vaccination outcomes - Join us for round 3!

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Abstract: Systems vaccinology studies have identified factors affecting individual vaccine responses, but comparing these findings is challenging due to varying study designs. To address this lack of reproducibility, we established a community resource, Computational Models of Immunity to Pertussis Booster vaccinations (CMI-PB), to compare Bordetella pertussis booster responses. CMI-PB (at https://www.cmi-pb.org/) provides experimental data for the explicit purpose of model evaluation, which is performed through a series of data releases and associated contests. We report here on our experiences from the first two rounds of prediction contests. The first 'dry-run' contest involved CMI-PB consortium members forming teams using different models to answer the contest questions. The second 'invited' contest included a select group of scientists from the broader community who have previously published in this area. In each contest, we assessed over 45 computational models that applied various methodologies including classification-based techniques, such as naive Bayes and random forest, regression-based approaches like elastic net, and various other strategies encompassing multi-omics integration, gene signature analysis, and module scoring. We found that handling baseline information on the target variables and data pre-processing is crucial for newly trained models. Overall, multiple co-inertia analyses gave the best results of the tested modeling approaches in both contests. We here present results from these contests, and invite the wider community to participate in round 3, a public prediction contest that will be launching in August 2024.

Introduction:

- Systems vaccinology studies have generated large datasets on the immune states of individuals before and after vaccination, and have identified factors that drive differences in individual vaccine responses.
- However, it is challenging to identify how conclusions from one study can be generalized across other studies given the differences in study design.
- We aim to address this lack of reproducibility by establishing a community resource and engaging the research community through open prediction challenges, which allow for the development and comparison of models that predict the immune response of human booster vaccinations for *Bordetella pertussis*.
- Here, we report on the outcome of the first challenge, an "internal dry run", where all teams involved in making predictions were part of the CMI-PB grant.
- We discuss here the challenges encountered in data sharing, formulating prediction questions, and interpretating results from different prediction models, and determining which factors contributed to such predictions. These results inform us in the design of the next prediction contest, which will open to community participation in August 2024.

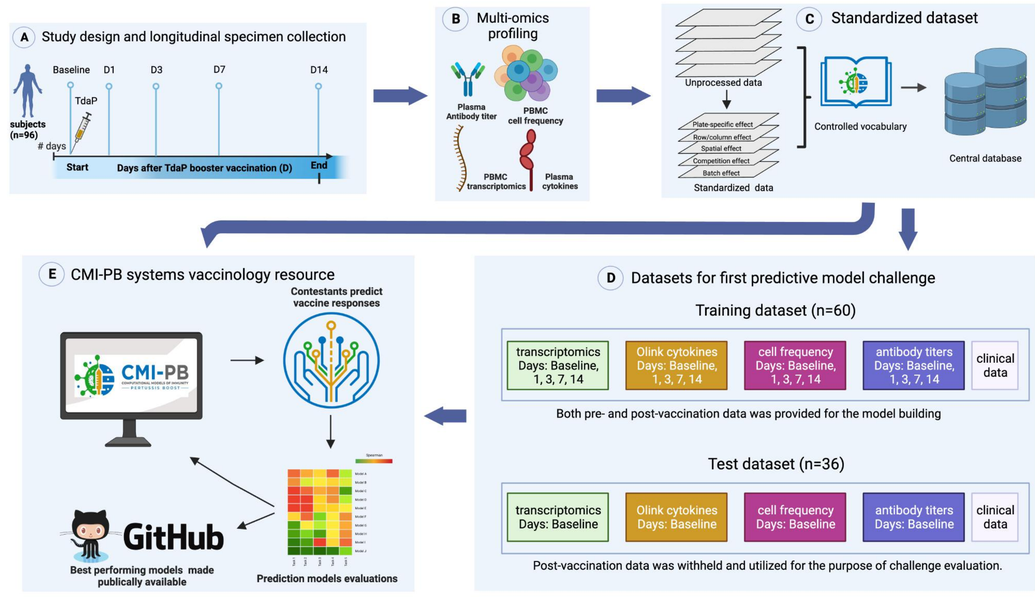
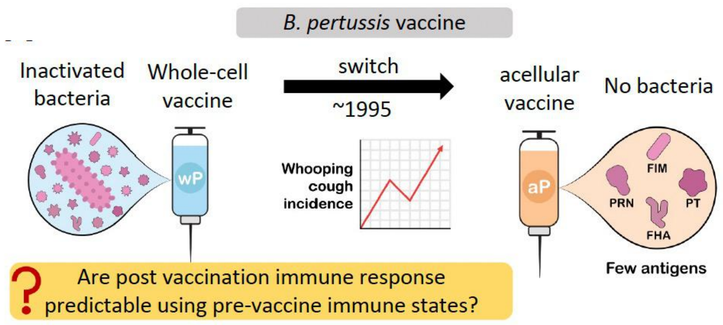


Fig 1: Outline for establishing the CMI-PB resource and prediction contest.

Methods and Results:



	Annual prediction challenge title	Contestants	Number of subjects		Current status
			Training dataset	Test dataset	
1	**First Challenge:** Internal dry run	CMI-PB consortium	60 (28 aP + 32 wP)	36 (19 aP + 17 wP)	Concluded in May 2022
2	**Second Challenge:** Invited challenge	Invited contestants	96 (47 aP + 49 wP)	23 (13 aP + 10 wP)	Concluded in January 2024
3	**Third Challenge:** Open Challenge	Public	119 (60 aP + 59 wP)	54 (27 aP + 27 wP)	Will be announced in August 2024

Table 1: Past and future CMI-PB annual prediction challenges.

List of prediction tasks

Task 1) Antibody levels task: Rank the individuals by IgG antibody titers against pertussis toxin that we detect in plasma 14 days post booster vaccinations.
Task 2) Cell frequency task: Rank the individuals by predicted frequency of Monocytes on day 1 after vaccination.
Task 3) Gene expression task: Rank the individuals based on their predicted CCL3 expression at day 3 post booster vaccination.

Models from the 1st Prediction Contest

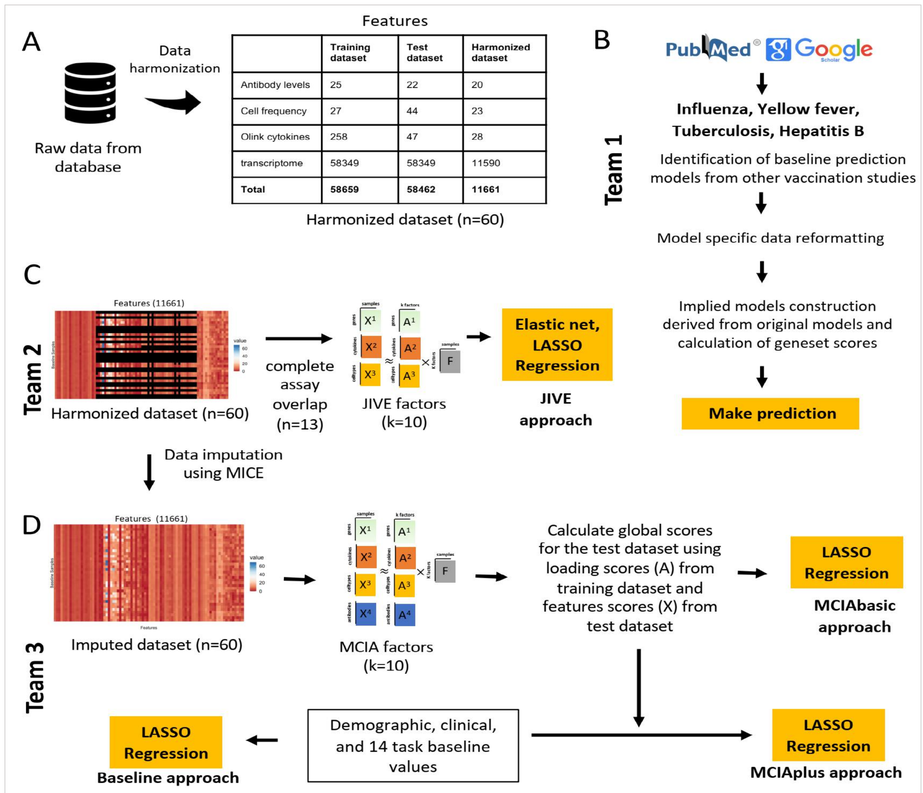


Fig 2: Data processing, computable matrices, and prediction model generation. Three teams employed their own methods for the development of models.

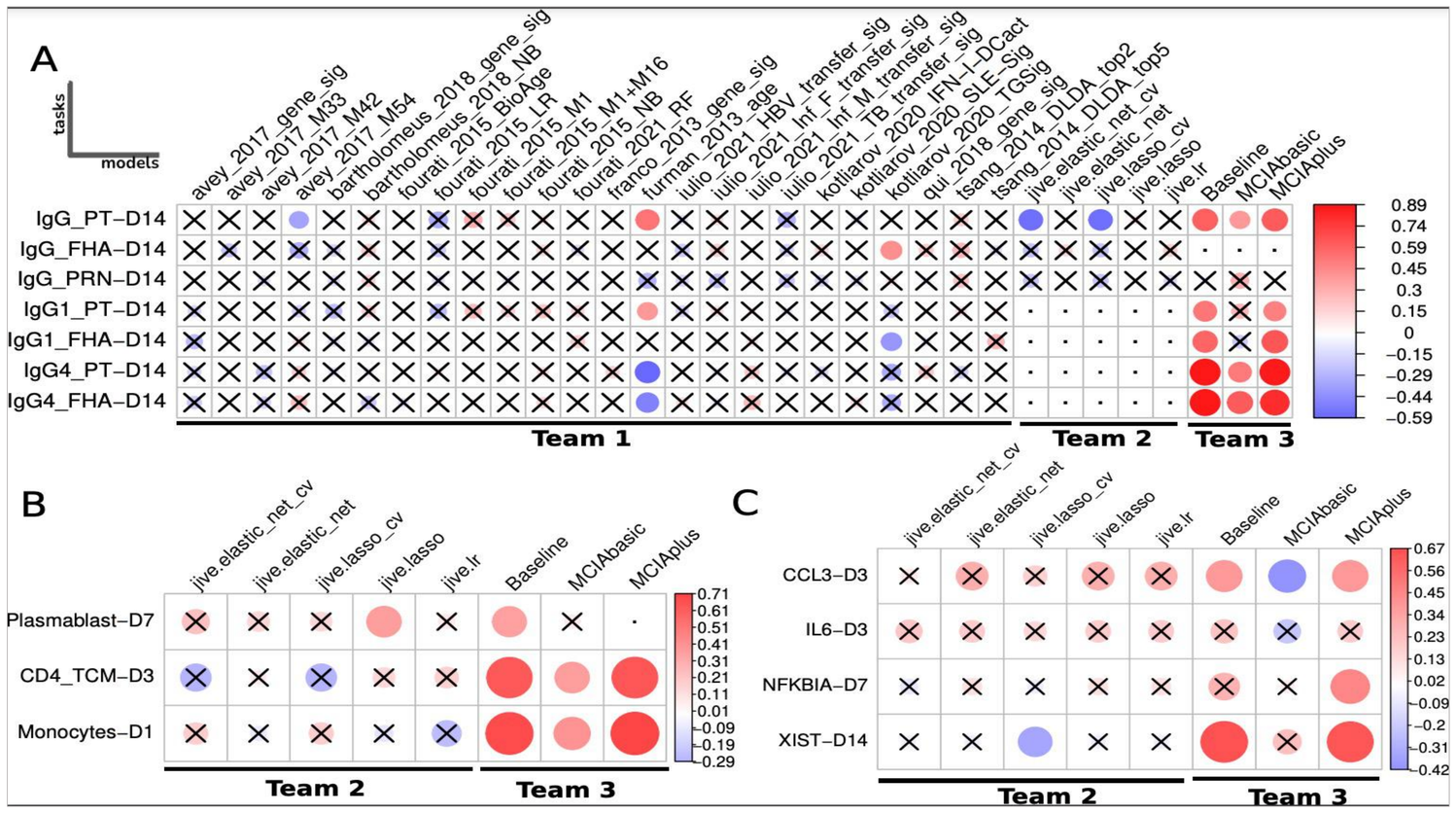


Fig 3: Evaluation of the prediction models submitted for the first CMI-PB challenge. Model evaluation was performed using Spearman's correlation coefficient between predicted ranks by a contestant and actual rank for each A) Antibody titers, B) Immune cell frequencies and C) transcriptomics tasks.

Conclusion:

- Over 20 models adopted from the literature were tested, but only one was predictive, and was based on age alone. The performance of new models built using CMI-PB training data was much better, but varied significantly based on the choice of pre-vaccination features used and the model-building strategy. This suggests that previously published models developed for other vaccines do not generalize well to Pertussis Booster vaccination.
- We believe that this collaborative and innovative approach will create a hub for immunologists to push for novel models of immunity against Tdap boost. We expect the resulting models will also be relevant for other vaccinology studies. Contestants from the research community who are interested in participating are encouraged to contact us via pshinde@lji.org and scan the QR to sign up to receive upcoming contest information.

References:

[1] Shinde, Pramod, et al. "A systems vaccinology resource to develop and test computational models of immunity." Cell Reports Methods, March 2024 (Featured on Journal cover page).

[2] da Silva Antunes, Ricardo, et al. "A system-view of B. pertussis booster vaccine responses in adults primed with whole-cell versus acellular vaccine in infancy." *JCI insight* (2021).

Acknowledgements:

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Scan to register for the Prediction Challenge or visit our website at www.cmi-pb.org

CMI-PB COMPUTATIONAL MODELS OF IMMUNITY PERTUSSIS BOOST